GLOBAL ENERGY & EMISSIONS
ANALYSIS REPORT
1990 – 2020

Submitted By
Prathmesh Mane | Tejas Mandalik | Rajwardhan Marne | Pratidnya Magar

1. Executive Summary

This report presents a comprehensive analysis of global energy production, consumption, and CO_2 emissions trends from 1990 to 2020, based on data sourced from the IEA / UNESCO Energy Statistics dataset. The dataset encompasses 44 countries across 7 global regions, covering 1,364 observations and 18 numeric indicators over a 31-year period.

The analysis reveals significant growth in CO_2 emissions driven primarily by rapid industrialization in Asia & Pacific and North America, alongside encouraging trends in renewable energy adoption, particularly from 2010 onwards. Wind and solar generation grew from negligible levels in 1990 to nearly 10% of electricity production by 2020, representing a structural shift in the global energy mix.

Dataset at a Glance

1,364	44	550.4	24.1%
observations	across 7 regions	$MtCO_2$ mean	of electricity
Total Records	**Countries**	**Avg CO_2**	**Avg Renewables**

2. Dataset Overview & Methodology

The dataset was compiled from IEA and UNESCO Energy Statistics databases, providing a robust longitudinal perspective on energy dynamics across developed and developing economies. The study covers the full spectrum of energy production and consumption metrics, including fossil fuel dominance and the progressive rise of renewable sources.

2.1 Data Source & Coverage

Parameter	Details
Source	IEA / UNESCO Energy Statistics
Countries	44 countries spanning all major economies

Regions	Africa, Arab States, Asia & Pacific, Europe, Middle East, North America, South/Latin America
Time Span	1990 to 2020 (31 years)
Total Observations	1,364
Numeric Indicators	18 indicators covering production, consumption, emissions, and renewables
Missing Data	Natural gas prod. (4.6%), Coal prod. (6.8%), Wind/Solar share (2.3%)

3. Regional Energy & Emissions Analysis

The regional breakdown reveals stark disparities in energy profiles, emissions intensity, and renewable adoption. North America leads in absolute CO_2 emissions, while South/Latin America demonstrates the highest renewable energy share, driven by Brazil's extensive hydropower infrastructure.

Region	CO_2 ($MtCO_2$)	Renewables %	Wind & Solar %	Electricity (TWh)	Trend
Africa	208.6	16.0%	0.37%	125.3	Moderate growth
Arab States	89.2	0.7%	0.11%	44.0	Low renewables
Asia & Pacific	852.0	16.7%	2.69%	536.9	Rapid expansion
Europe	341.5	25.0%	4.07%	259.8	Strong transition
Middle East	218.2	4.4%	0.22%	122.1	Fossil dependent
North America	2,910.5	37.0%	2.23%	2,313.7	Highest emitter
South/Latin America	184.8	55.2%	1.39%	160.9	Renewables leader

Key Insight: North America accounts for the highest absolute CO_2 emissions (2,910.5 $MtCO_2$ average), while South/Latin America achieves the highest renewable share (55.2%) due to extensive hydropower use in Brazil, Colombia, and Venezuela.

4. Temporal Trends (1990–2020)

Tracking energy and emissions metrics over three decades reveals distinct phases: a gradual rise in the 1990s, acceleration in the 2000s driven by Asian industrialization, a peak in the mid-2010s, and the beginning of a structural renewable transition by 2020.

4.1 CO_2 Emissions Trend

Year	CO_2 ($MtCO_2$)	Renewables %	Wind & Solar %	Electricity (TWh)	Energy Cons. (Mtoe)
1990	420.4	23.9%	0.46%	245.8	173.5
1995	440.1	24.4%	0.48%	276.5	183.5
2000	477.7	22.6%	0.65%	321.2	199.0

2005	557.1	22.3%	1.14%	380.5	227.9
2010	626.2	23.3%	2.51%	447.6	254.4
2015	662.1	25.3%	5.65%	505.1	268.0
2020	645.1	30.0%	9.77%	554.9	272.0

Notable: CO_2 peaked around 2015 at 662 $MtCO_2$, then declined to 645 $MtCO_2$ by 2020 — partly attributable to the COVID-19 pandemic and accelerating renewable deployment. Wind & Solar grew 21x from 0.46% (1990) to 9.77% (2020).

5. Country-Level Analysis

5.1 Top 10 Countries by CO_2 Emissions (Avg, $MtCO_2$)

Rank	Country	CO_2 ($MtCO_2$)	Renewables %	Electricity (TWh)	Elec. Consump. (TWh)
1	**China**	5,724.5	19.5%	3,230.4	2,785.2
2	**United States**	5,297.0	11.8%	4,025.5	3,587.2
3	**Russia**	1,641.5	17.8%	986.5	802.0
4	**India**	1,292.7	17.5%	833.1	609.7
5	**Japan**	1,122.0	12.0%	1,045.2	953.3
6	**Germany**	809.1	15.8%	596.1	512.3
7	**Canada**	524.1	62.1%	601.9	515.7
8	**South Korea**	490.8	2.9%	373.2	342.0
9	**United Kingdom**	487.5	10.8%	357.4	321.6
10	**Iran**	418.6	6.9%	182.3	146.9

6. Descriptive Statistics — Key Indicators

The following table presents a statistical summary of the core energy and emissions indicators across all 1,364 observations, providing insight into distribution, variability, and outliers within the dataset.

Indicator	Mean	Std Dev	Min	25th %ile	Median	75th %ile	Max
CO_2 Emissions ($MtCO_2$)	550.4	1,213.1	7.6	99.9	221.0	423.8	9,716.8
Renewables Share (%)	24.1	25.5	0.0	6.1	14.9	33.3	99.8
Electricity Prod. (TWh)	388.6	834.8	10.8	70.0	146.8	308.4	7,797.6
Energy Consumption (Mtoe)	226.0	445.2	2.6	46.6	93.8	195.8	3,381.4

Wind & Solar Share (%)	2.54	5.01	0.0	0.003	0.17	2.73	32.1
Natural Gas Prod. (bcm)	61.3	133.6	0.0	4.4	21.1	55.0	959.9
Crude Oil Prod. (Mt)	74.1	119.7	0.0	1.6	25.9	106.4	748.2

7. Key Findings & Insights

7.1 Emissions Growth

- Global average CO_2 emissions grew by 53% from 420.4 $MtCO_2$ in 1990 to 645.1 $MtCO_2$ in 2020.
- China and the United States together account for the majority of global CO_2 emissions, with China averaging 5,724.5 $MtCO_2$ and the US averaging 5,297.0 $MtCO_2$.
- CO_2 emissions peaked in 2015 at approximately 662 $MtCO_2$ and showed a decline by 2020, influenced by COVID-19 lockdowns and renewable energy expansion.

7.2 Renewable Energy Transition

- The share of renewables in electricity production grew from 23.9% in 1990 to 30.0% in 2020.
- Wind and solar generation experienced the most dramatic growth, increasing from 0.46% (1990) to 9.77% (2020) — a 21x increase over 30 years.
- South/Latin America leads globally in renewable share (55.2%), primarily driven by hydropower. Norway holds the highest single-country renewable share at approximately 98.7%.
- Europe demonstrated the strongest wind & solar growth trajectory, averaging 4.07% across the period.

7.3 Energy Intensity Improvements

- Energy intensity of GDP (koe/$15p) showed a consistent decline from 0.176 in 1990 to 0.117 in 2020, indicating improving energy efficiency across economies.
- CO_2 intensity at constant purchasing power parities also declined from 0.416 (1990) to 0.253 (2020), reflecting a decoupling of economic growth from emissions.

7.4 Regional Disparities

- North America produces and consumes the most energy in absolute terms, with electricity production averaging 2,313.7 TWh — more than four times the global average.
- Arab States have the lowest renewable energy share (0.7%), underscoring dependence on fossil fuel exports for domestic energy supply.
- Africa, despite substantial energy production, has a relatively low per-capita electricity consumption, indicating ongoing energy access challenges.

8. Conclusions & Recommendations

The three-decade analysis of global energy and emissions data presents a nuanced picture: while CO_2 emissions have grown substantially, the rate of growth has slowed and the energy mix is

transitioning. The data underscores both the scale of the challenge and the accelerating pace of the renewable energy revolution.

8.1 Policy Implications

- Accelerated investment in wind and solar infrastructure is critical, given the 21x growth in wind/solar share demonstrates the scalability of these technologies.
- Energy efficiency programs remain essential; the declining energy intensity of GDP confirms that economic growth can be achieved with lower energy inputs.
- Developing regions, particularly Africa and Arab States, require targeted support for renewable deployment to leapfrog fossil fuel dependency.
- International cooperation frameworks must address the asymmetry between high-emission economies (China, USA) and smaller contributors disproportionately affected by climate impacts.

8.2 Future Outlook

Based on observed trends, renewable energy's share of electricity generation is expected to continue accelerating through 2030 and beyond. The declining CO_2 trajectory from 2015 to 2020, if sustained, aligns with intermediate targets under the Paris Agreement. However, absolute emissions levels remain far above those required for a 1.5°C pathway, necessitating deeper structural changes in energy systems globally.

9. References & Data Sources

- IEA / UNESCO Energy Statistics Database — Global Energy Dataset (1990–2020)
- International Energy Agency (IEA): World Energy Outlook series
- GeeksforGeeks: C Program for Hospital Management System — https://www.geeksforgeeks.org/c/c-program-for-hospital-management-system/
- GitHub Repository: EEL Final Project — https://github.com/pratidnya102-png/pratidnya-/
- Scribd Reference Document — https://www.scribd.com/document/431969780/mine

Declaration of Authorship

We hereby declare that this report is our original work, prepared as part of the EEL Final Project.

Prathmesh Mane | Tejas Mandalik | Rajwardhan Marne | Pratidnya Magar